PH 113
Physics

Nuclear Fission and Fusion

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Binding Energy

A binding energy is generally the energy required to disassemble a whole system into separate parts. It is known the sum of separate parts has typically a higher potential energy than a bound system, therefore the bound system is more stable. A creation of bound system is often accompanied by subsequent energy release.

Nuclear Binding Energy

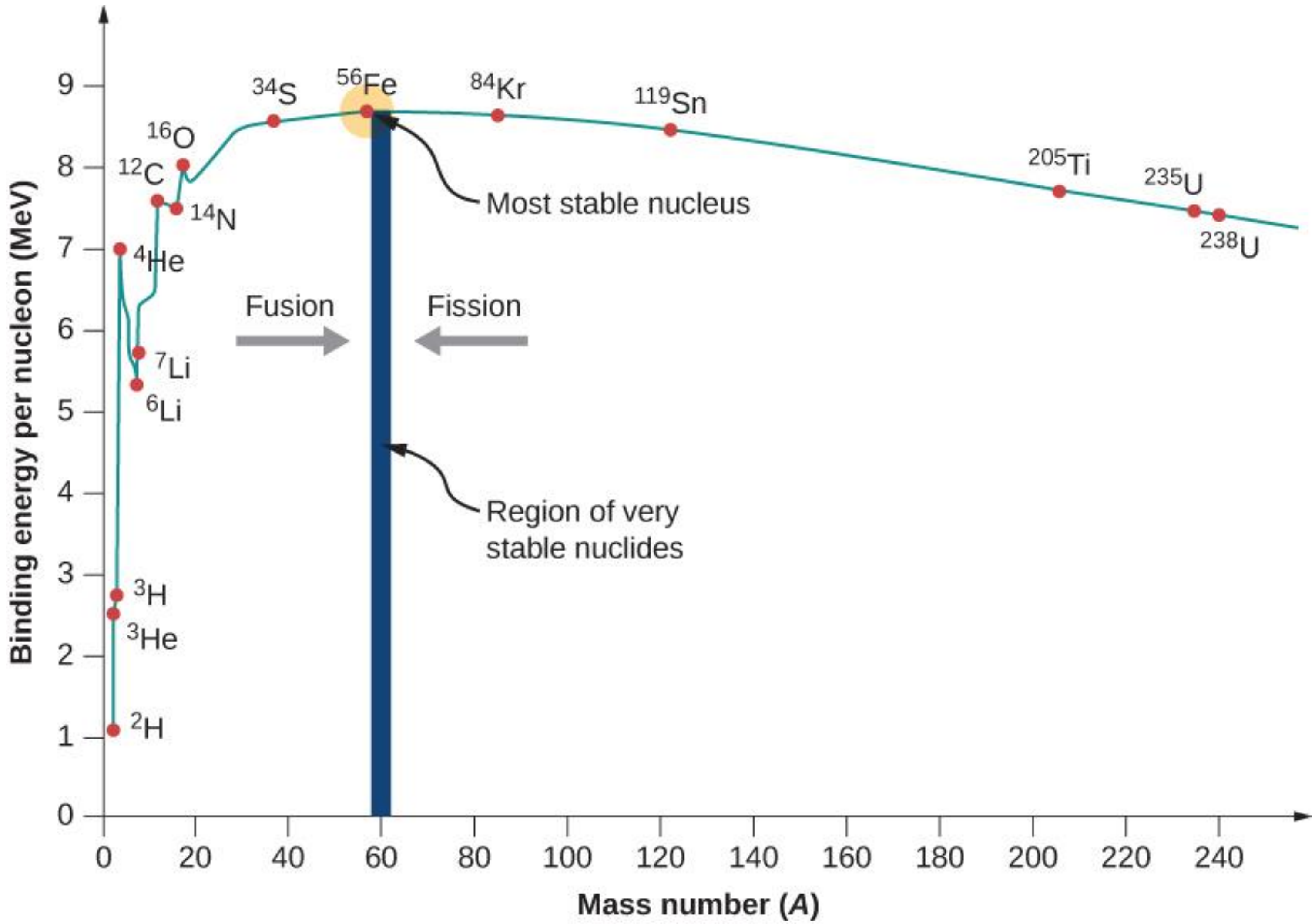
The component parts of nuclei are neutrons and protons, which are collectively called nucleons. **The mass of a nucleus is always less than the sum masses of the constituent protons and neutrons when separated**. The difference is a measure of the nuclear binding energy which holds the nucleus together. According to the Einstein relationship ($E=m.c^2$) this binding energy is proportional to this mass difference and it is known as **the mass defect**.

Nuclear Binding Curve

If the splitting releases energy and the fusion releases the energy, so where is the breaking point? For understanding this issue it is better to relate the binding energy to one nucleon, to obtain **nuclear binding curve**. The binding energy per one nucleon is not linear. There is a peak in the binding energy curve in the region of stability near **iron** and this means that either the breakup of heavier nuclei than iron or the combining of lighter nuclei than iron will yield energy.

Nuclear Binding Curve

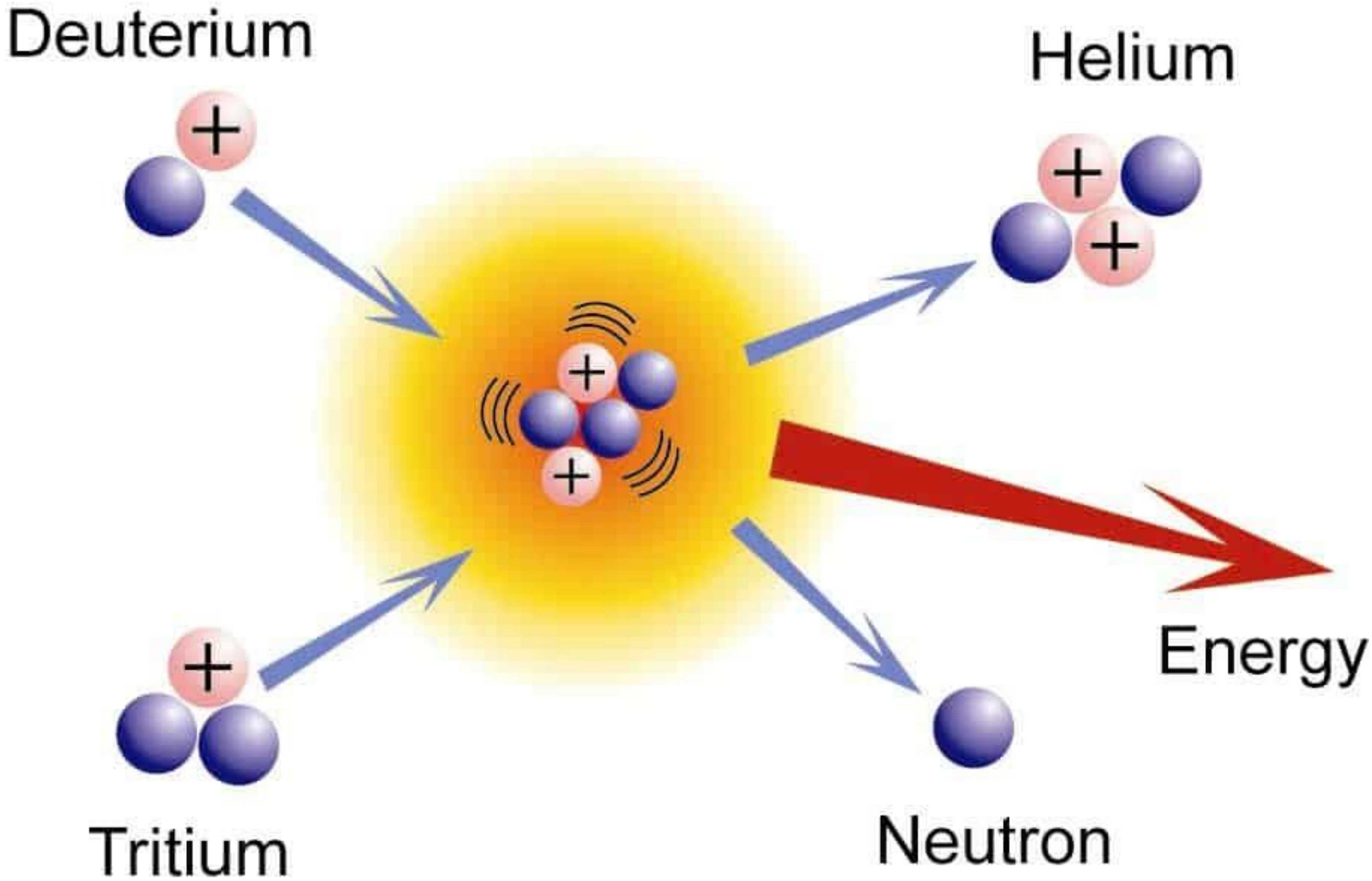
The reason the trend reverses after iron peak is the growing positive charge of the nuclei. The electric force has greater range than strong nuclear force. While the strong nuclear force binds only close neighbors the electric force of each proton repels the other protons.



Binding energy per nucleon (MeV)
Mass number (A)
2H
3He
3H
4He
6Li
7Li
12C
14N
16O
34S
56Fe
84Kr
119Sn
205Ti
235U
238U
Most stable nucleus
Fusion
Fission
Region of very stable nuclides
0
1
2
3
4
5
6
7
8
9
0
20
40
60
80
100
120
140
160
180
200
220
240

Nuclear Fusion

In nuclear physics, **nuclear fusion** is a nuclear reaction in which two or more atomic nuclei collide at a very high energy and fuse together into a new nucleus, e.g. helium. If light nuclei are forced together, they will fuse with a yield of energy because the mass of the combination will be less than the sum of the masses of the individual nuclei. If the combined nuclear mass is less than that of iron at the peak of the **binding energy curve**, then the nuclear particles will be more tightly bound than they were in the lighter nuclei, and that decrease in mass comes off in the form of energy according to the Albert Einstein relationship.


Deuterium
Helium
Energy
Tritium
Neutron

Nuclear Fusion

Fusion reactions have an energy density many times greater than nuclear fission and fusion reactions are themselves millions of times more energetic than chemical reactions.

The fusion power offers the opportunity of an almost inexhaustible source of energy for future, but it the fusion technology presents a real scientific and engineering challenges.

Deuterium-Tritium Fusion

The fusion reaction of deuterium and tritium is particularly interesting because of its potential of providing energy for the future.

3T (d, n) 4He

The reaction yields ~17 MeV of energy per reaction.

Nuclear Fission

Nuclear fission refers to the splitting of an atomic nucleus into two or more lighter nuclei. This process can occur through a nuclear reaction or through radioactive decay. Nuclear fission reactions often release a large amount of energy, which is accompanied by the emission of neutrons and gamma rays (photons holding huge amounts of energy, enough to knock electrons out of atoms).

Types of Nuclear Fission

- **Spontaneous Fission:** Nuclei consisting of protons, electrostatic force of repulsion between them exceeds the nuclear binding force hence nuclei undergoes spontaneous fission
- **Induced Fission:** When neutron is bombarded on heavy nucleus it makes nucleus unstable this unstable nucleus undergoes fission and split into two fragment. Such fission is called as induced fission.